

OptimalControl.jl: A Julia package for modeling and solving optimal control problems with ODEs

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Submitted: 01 January 1970

Published: unpublished

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Summary

OptimalControl.jl (Caillau, Cots, Gergaud, Martinon, & Sed, 2024) is a Julia (Bezanson et al., 2017) package for modeling and solving optimal control problems governed by ordinary differential equations (ODEs). As the core of the [control-toolbox ecosystem](#), it provides a unified framework that supports both direct and indirect solution methods with applications spanning aerospace, medical imaging, epidemiology, and quantum control.

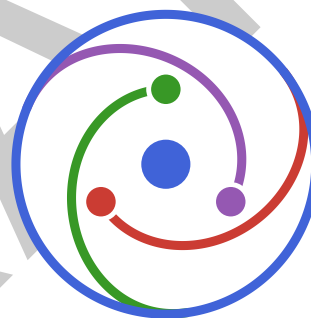


Figure 1: Logo of OptimalControl.jl.

The package features an expressive domain-specific language (DSL) built around the `@def` macro, enabling users to define control problems using notation that closely resembles standard mathematical formulations. Problems are solved through direct transcription, converting the continuous problem into a nonlinear program (NLP) using discretization schemes including Euler, trapezoidal, midpoint, and high-order Gauss-Legendre collocation. Alternatively, indirect shooting methods based on Pontryagin's Maximum Principle can be employed. The architecture relies on a modeler-solver separation that provides a modular and extensible foundation, enabling GPU execution (currently limited to NVIDIA hardware) with minimal user intervention alongside standard CPU solvers.

The ecosystem includes extensive [tutorial resources](#), a benchmark problem collection ([OptimalControlProblems.jl](#) (Caillau, Cots, Gergaud, & Martinon, 2024) with formulations in OptimalControl DSL and [JuMP](#) (Lubin et al., 2023)), and performance comparison tools ([CTBenchmarks.jl](#)). Integration with Julia's ecosystem enables access to state-of-the-art tools: NLP solvers [IPOPT](#) (Wächter & Biegler, 2006) and [MadNLP.jl](#) (Shin et al., 2021, 2024), automatic differentiation and NLP modeling through JuliaSmoothOptimizers' [ADNLPModels](#) (Migot et al., 2023), GPU acceleration via [ExaModels.jl](#) (Shin et al., 2023), numerical integration from SciML's [DifferentialEquations.jl](#) (Rackauckas & Nie, 2017), and visualization

31 through [Plots.jl](#) ([Christ et al., 2023](#)).

32 Problem Class

33 The package currently targets single-phase problems governed by explicit ODEs, written in
34 Bolza form,

$$g(t_0, x(t_0), t_f, x(t_f), v) + \int_{t_0}^{t_f} f^0(t, x(t), u(t), v) dt \rightarrow \min,$$

35 subject to the dynamics $\dot{x}(t) = f(t, x(t), u(t), v)$, where $x(t) \in \mathbb{R}^n$ is the state, $u(t) \in \mathbb{R}^m$
36 the control, and $v \in \mathbb{R}^q$ an optimisation variable, that is a finite-dimensional parameter
37 optimised together with the trajectory. Mayer ($f^0 = 0$) and Lagrange ($g = 0$) costs are
38 the corresponding special cases, and maximisation is also supported. Both autonomous and
39 non-autonomous dynamics are handled, the dependences being detected by the parser. The
40 time bounds t_0 and t_f may be fixed or be components of v , which covers free initial and / or
41 final time, in particular minimum time problems. Admissible constraints are of five types —
42 boundary, variable, control, state, and mixed (control and state), the last three being *path*
43 constraints, imposed at all times — and may be equalities or one- or two-sided inequalities,
44 either box constraints on state, control and variable (for which dedicated bounds are passed
45 to the NLP solver) or nonlinear. The variable may appear in any of them. The current
46 implementation is restricted to *single-phase* problems with *explicit* ODE dynamics: multiphase
47 problems, differential-algebraic equations, and hybrid or nonsmooth systems are not supported.
48 Optimization of PDE or stochastic systems is out of scope as well (see the design trade-offs
49 below).

50 Statement of Need

51 OptimalControl.jl is, to our knowledge, the only Julia package that unifies both direct and
52 indirect methods for optimal control within a single, coherent framework.

53 This unified approach addresses a gap in the current optimal control software landscape.
54 Existing tools are fragmented across programming languages and paradigms. Legacy packages
55 such as [HamPath](#) ([Caillaud et al., 2012](#)), [NutoPy](#) ([Caillaud et al., 2022](#)), and [COTCOT](#) ([Bonnard et al., 2007](#))
56 implement sophisticated indirect methods but rely on Fortran implementations with
57 MATLAB or Python interfaces, leading to complex and less extensible workflows. Proprietary
58 solvers like [GPOPS-II](#) ([Patterson & Rao, 2014](#)) limit transparency and reproducibility. Open-
59 source tools such as [BOCOP](#) ([Bonnans et al., 2017](#)), [ACADO](#) ([Houska et al., 2011](#)), and
60 [acados](#) ([Verschuere et al., 2021](#)) provide valuable direct method implementations but are not
61 natively integrated into a modern, high-level scientific computing ecosystem. Python-based
62 tools such as [Dymos](#) ([Falck et al., 2021](#)), [Pyomo.DAE](#) ([Nicholson et al., 2018](#)), [do-mpc](#) ([Fiedler et al., 2023](#)),
63 and [GEKKO](#) ([Beal et al., 2018](#)) offer complementary capabilities but are similarly
64 limited to direct transcription approaches within the Python ecosystem.

65 Within Julia, most existing packages target specific domains: [RobustAndOptimalControl.jl](#) for
66 linear systems, [QuantumControl.jl](#) ([Machnes et al., 2025](#)) and [Piccolo.jl](#) for quantum optimal
67 control, [DirectTrajectoryOptimization.jl](#) for trajectory problems, [LinearMPC.jl](#) ([Arnström et al., 2022](#))
68 and [ModelPredictiveControl.jl](#) ([Gagnon et al., 2024](#)) for model predictive control,
69 and [InfiniteOpt.jl](#) ([Pulsipher et al., 2022](#)) for infinite-dimensional optimization. In contrast to
70 [InfiniteOpt.jl](#), which is designed as an extension of [JuMP.jl](#), our package provides a modeler-
71 agnostic approach, accepting general Julia code and leveraging various optimization solvers.
72 Moreover, contributing to general-purpose NLP modeling frameworks like [JuMP](#) or [CasADi](#)
73 ([Andersson et al., 2019](#)) would not address the specific needs of optimal control: these tools

74 lack native support for shooting methods based on Pontryagin's Maximum Principle, which
75 heavily rely on differential geometric primitives.

76 OptimalControl.jl fills this gap by providing a DSL that matches mathematical notation, multiple
77 discretization schemes and shooting methods, with planned support for homotopy continuation
78 methods. The modeler-solver separation enables complementary use with [InfiniteOpt.jl](#) (JuMP-
79 based) through alternative NLP modeling backends (currently [ADNLPModels](#), [ExaModels](#))
80 and solvers. GPU acceleration and an ecosystem with domain-specific applications, tutorials,
81 and benchmarking tools complete the offering. Target users include researchers and engineers
82 working in optimal control, control theorists developing new algorithms, and students learning
83 optimal control through interactive tutorials.

84 State of the Field

85 OptimalControl.jl requires Julia version 1.10 or later and is registered in the Julia General
86 registry, enabling straightforward installation via `Pkg.add("OptimalControl")`. We compare
87 it below with existing software.

- 88 ■ **Legacy tools (COTCOT, HamPath, NutoPy):** These Fortran packages excel at indirect
89 methods and homotopy continuation but require multi-language setup (Fortran plus
90 MATLAB / Python). OptimalControl.jl provides both direct and indirect methods in
91 pure Julia with straightforward installation via package manager.
- 92 ■ **Direct-method tools (BOCOP, ACADO, GPOPS-II, acados, nosnoc):** Strong direct-
93 method implementations: [GPOPS-II](#) delivers mature methods with MATLAB and C++
94 implementations but is proprietary; [acados](#) targets real-time MPC on embedded systems;
95 [nosnoc](#) ([Nurkanović & Diehl, 2022](#)) specializes in nonsmooth optimal control; [CasADi](#),
96 used as symbolic backend by several of these tools, is a general NLP modeler rather
97 than an optimal control solver. OptimalControl.jl offers an open-source alternative with
98 expressive DSL, native Julia ecosystem integration, GPU support, and unified direct and
99 indirect approaches.
- 100 ■ **Julia packages:** [RobustAndOptimalControl.jl](#) targets linear systems; [QuantumControl.jl](#),
101 [Piccolo.jl](#) and [DirectTrajectoryOptimization.jl](#) serve specific domains; [LinearMPC.jl](#) and
102 [ModelPredictiveControl.jl](#) focus on model predictive control. [InfiniteOpt.jl](#) addresses a
103 very rich range of problems, including optimization on PDEs or with chance constraints,
104 focusing on direct transcription methods. OptimalControl.jl supports both CPU and
105 GPU execution (currently NVIDIA-only, via [ExaModels.jl](#) + [MadNLP.jl](#)), and adds
106 tools to do shooting in a unified framework plus systematic benchmarking through
107 [OptimalControlProblems.jl](#) and [CTBenchmarks.jl](#).

108 Illustrative Example

109 A self-contained illustrative example is provided as a companion repository ([Caillaud et al., 2026](#)),
110 archived on Zenodo. It combines direct and indirect solution approaches for a constrained
111 energy minimization problem: a direct method on a coarse grid identifies the three-arc
112 structure (unconstrained–constrained–unconstrained) and initializes a shooting method based
113 on Pontryagin's Maximum Principle, which then converges to arbitrary precision.

114 Software Design

115 The package architecture balances expressiveness, performance, and extensibility through
116 modular design. The core is organized across internal packages within the [control-toolbox](#)
117 [organization](#), each hosted as an independent Julia package and registered in the General
118 registry: [CTBase.jl](#) defines the core types (optimal control models, solutions, initial guesses)

and their accessors; [CTParser.jl](#) implements the `@def` DSL macro; [CTDirect.jl](#) handles direct transcription and NLP interfacing; and [CTFlows.jl](#) provides Hamiltonian flows and shooting for indirect methods. [OptimalControl.jl](#) re-exports these packages as a unified entry point. Contributors interested in a specific functionality can work directly on the relevant sub-package, each of which has its own documentation, tests, and continuous integration. The delegation of specific responsibilities is as follows:

- **DSL parsing:** [MLStyle.jl](#) (Thautwarm & contributors, 2023) enables pattern matching on abstract syntax trees generated from `@def` macro invocations. This design choice separates problem specification from solution methods, and allows a syntax as close as possible to the mathematical description of the problem.
- **Problem models and solutions:** Structured types represent optimal control models, solutions, and initial guesses. Each type provides textual visualization, accessor methods for introspection, and visual plotting capabilities for solutions. Extensible abstractions enable addition of new problem classes without modifying core code.
- **Discretization:** Direct transcription converts continuous problems into NLPs. Supporting multiple discretization schemes (Euler, midpoint, trapezoidal, Gauss-Legendre) required careful abstraction to share transcription logic while allowing scheme-specific implementations.
- **Modelers and solvers:** We chose a clear modeler-solver separation: discretization produces [ADNLPModels](#) (able to deal with arbitrary user-defined functions) or [ExaModels](#) instances compatible with multiple NLP solvers ([IPOPT](#) via [NLPModelsIpopt.jl](#) (Orban et al., 2023), [Knitro](#) (Byrd et al., 2006), [MadNLP](#), and [Uno](#) (Vanaret & Leyffer, 2026)). These NLP solvers rely on linear solvers such as [MUMPS.jl](#) (Amestoy et al., 2001, 2019; Montoisin et al., 2024) for CPU computations or [CUDSS.jl](#) (NVIDIA Corporation, 2025) for GPU acceleration with [MadNLP](#). Automatic differentiation via [ForwardDiff.jl](#) (Revels et al., 2016) and [DifferentiationInterface.jl](#) (Dalle & Hill, 2026) avoids manual derivative coding.
- **Indirect methods:** Hamiltonian flows integrate with [DifferentialEquations.jl](#) to access adaptive stepping, event handling, and multiple ODE solvers without reimplementation. Shooting methods rely on [NonlinearSolve.jl](#) (Pal et al., 2025) or [MINPACK.jl](#) (Moré et al., 1980) for root-finding. Future extensions will incorporate homotopy continuation methods leveraging bifurcation analysis tools like [BifurcationKit.jl](#) (Veltz, 2020). Currently, indirect methods are limited to Hamiltonian flows, where the feedback control law depends on both state and costate; support for non-Hamiltonian flows (open-loop or state-feedback control laws) is under active development.
- **Testing and quality assurance:** The multi-repository structure is matched by a layered testing strategy. Each sub-package is tested independently, combining unit tests, integration tests, and code-quality checks, while [OptimalControl.jl](#) adds strong end-to-end integration tests that solve complete optimal control problems by both direct and indirect methods. Continuous integration runs across Linux, macOS, and Windows, on both CPU and GPU (through a self-hosted CUDA runner), using reusable GitHub Actions workflows centralized in [CTActions](#) and shared across the ecosystem. Code coverage is tracked on Codecov, downstream packages and applications are guarded against regressions through dedicated breakage tests triggered on pull requests, and beta versions are distributed during development via a local registry, [ct-registry](#). Regression testing on complex optimal control instances is increasingly supported by [OptimalControlProblems.jl](#), which serves as a curated problem bank.

Key design trade-offs:

1. **DSL vs. programmatic API:** We prioritized a mathematical DSL to reduce cognitive load for domain experts, accepting increased parsing complexity handled internally. Currently we do not address optimization of PDE or stochastic systems.

- 170 2. **GPU acceleration strategy:** The modeler-solver separation enables a straightforward
171 CPU-to-GPU transition. Users select the [ExaModels](#) modeler with [MadNLP](#) and the
172 [CUDSS](#) linear solver for GPU execution, or simply append the `:gpu` token to the `solve` call.
173 This modular approach minimizes maintenance burden while enabling GPU performance
174 without reimplementing transcription logic. GPU support is currently limited to NVIDIA
175 hardware and to the [ExaModels](#) + [MadNLP](#) modeler-solver combination.
- 176 3. **Method coverage:** Supporting both direct and indirect approaches increases code
177 complexity but serves distinct user needs: direct methods for constrained problems with
178 many variables, indirect methods for theoretical analysis and smaller problems requiring
179 high accuracy. Both approaches rely on iterative solvers and may or may not converge,
180 e.g., depending on the initial guess. In the case of optimization solvers, the full output
181 status of the solver is returned allowing *a posteriori* analysis.

182 Research Impact Statement

183 OptimalControl.jl has been applied in published research across multiple domains. Independent
184 external adoption demonstrates the package's broader impact: Ferede et al. (2025) used it for
185 drone racing trajectory optimization, Caio & Silva (2025) for epidemiological models using
186 indirect methods, Morsky (2025) for vaccination strategies under social norms, Opmeer (2025)
187 for optimal harvesting of age-structured populations, and Evangelakos et al. (2025) for fast
188 charging protocols for quantum batteries.

189 Research by close collaborators further demonstrates the package's versatility across aerospace
190 (Herasimenka et al. (2026)), epidemiology (Bliman et al. (2026)), quantum control
191 (Beschastnyi et al. (2025)), microbial growth control (Innerarity Imizcoz et al. (2025)), and
192 theoretical optimal control (Bouali et al. (2025); Lutz & Privat (2025); Bayen et al. (2026);
193 Bonnard et al. (2026)).

194 The [control-toolbox organization](#) hosts domain-specific application packages built on
195 OptimalControl.jl: medical imaging optimization ([MagneticResonanceImaging.jl](#)), gene
196 regulatory networks ([PWLdynamics.jl](#)), spacecraft orbital transfers ([Kepler.jl](#)), epidemiological
197 modeling ([SIRcontrol.jl](#)), and variational calculus ([CalculusOfVariations.jl](#)).

198 The package also serves educational purposes through [Tutorials.jl](#), which covers topics from
199 linear-quadratic regulators to Model Predictive Control. These resources are used in academic
200 courses and workshops; the package was notably presented at JuliaCon 2023 ([Caillau et al., 2023](#)).
201 [OptimalControlProblems.jl](#) provides standardized test problems formulated in
202 both OptimalControl DSL and [JuMP](#), enabling systematic performance comparisons through
203 [CTBenchmarks.jl](#).

204 Development involves international collaborations with CNES (French space agency), Thales
205 Alenia Space, Inria, and CNRS. The package development began in September 2022, with
206 the first public release in February 2023. As of March 2026, OptimalControl.jl has published
207 more than 40 releases with contributions from multiple developers across the control-toolbox
208 ecosystem, demonstrating sustained community engagement.

209 AI Usage Disclosure

210 The core software implementation, algorithms, and architectural design of OptimalControl.jl
211 were developed by the authors. Generative AI tools—specifically Claude Sonnet (version 4.6)
212 and Claude Opus (version 4.8) by Anthropic—have been used to assist with code refactoring
213 and generation of unit and integration tests, always under human review. This paper was
214 drafted by the authors and subsequently revised with AI assistance for restructuring according
215 to JOSS format requirements and language editing. All technical content, examples, and
216 claims reflect the authors' work and judgment.

Acknowledgements

We thank the [control-toolbox community](#) and the broader Julia ecosystem for contributions that shaped the package design and educational resources. Development has been supported by partnerships with CNES, Thales Alenia Space, Inria, and CNRS. The software is distributed under the MIT license (see [LICENSE](#)). We welcome community contributions following the guidelines in [CONTRIBUTING.md](#).

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